

## Windows Login Brute Force Detection Example

### 7.1 Windows Login Records

In general, login activity is present in all successful or unsuccessful cyberattacks. Often, an attacker wants to log into the server to take over the system. To do so, they can perform a brute force attack or log in directly with the password. In both cases (successful login or unsuccessful login attempt), the log will be created.

Suppose an attacker logs into the server after a brute force attack. To better analyze the attacker's actions after entering the system, we need to find the login date. To do so, we must find "Event ID 4624: An account was successfully logged on."

Each event log has its own ID value. Filtering, analyzing and searching the log title is more difficult, so it is easy to use the ID value.

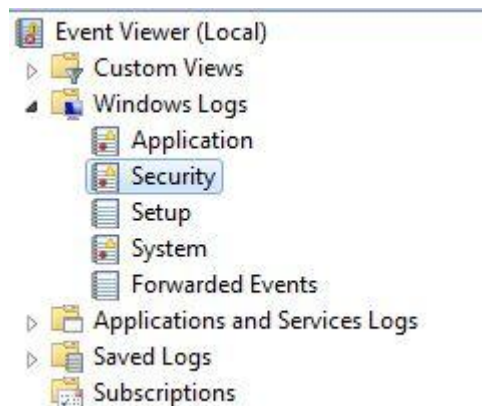
You can find the details of which Event ID value means what from the URL address below.

<https://www.ultimatewindowssecurity.com/securitylog/encyclopedia/default.aspx>

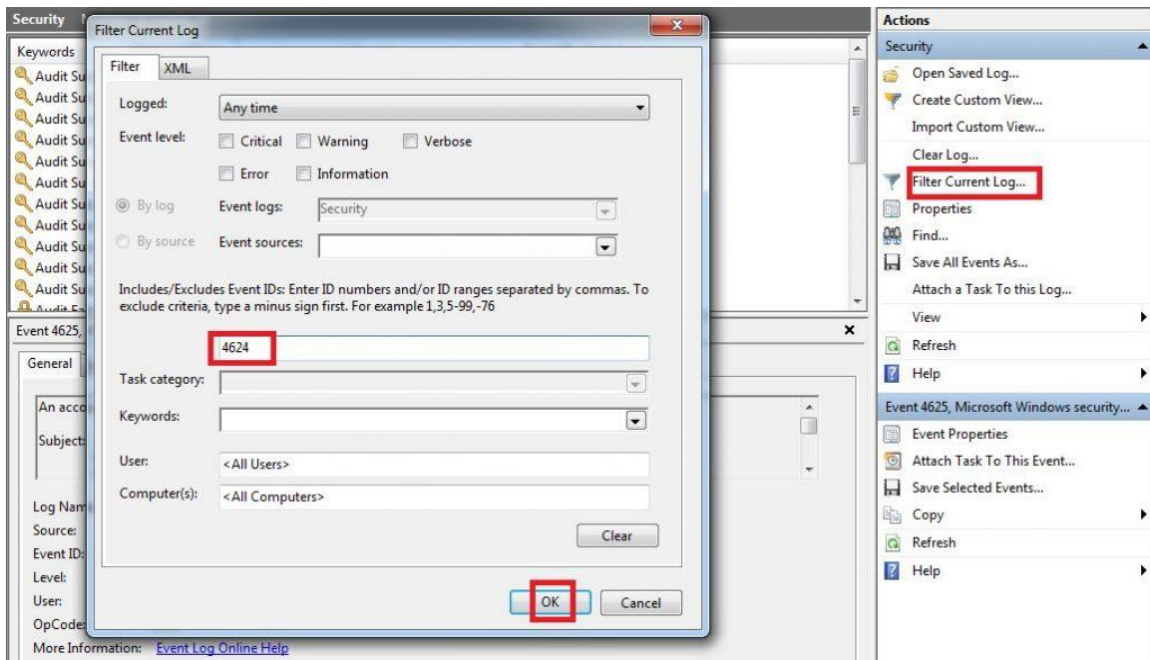
Log file for lesson:

[Log\\_File.zip](https://files-lid.s3.us-east-2.amazonaws.com/Log_File.zip) Pass=321 ([https://files-lid.s3.us-east-2.amazonaws.com/Log\\_File.zip](https://files-lid.s3.us-east-2.amazonaws.com/Log_File.zip))

To reach the result, we open the "Event Viewer" and select "Security" logs.



Then we create a filter for the "4624" Event ID.



And now we see that the number of logs has decreased significantly and we are only listing logs for successful login activities. Looking at the log details, we see that the user of “LetsDefendTest” first logged in at 23/02/2021 10:17 PM.

Security Number of events: 24

Filtered: Log: Security; Source: ; Event ID: 4624. Number of events: 3

| Keywords      | Date and Time         | Source            | Event ID | Task Category |
|---------------|-----------------------|-------------------|----------|---------------|
| Audit Success | 2/23/2021 10:17:31 PM | Microsoft Wind... | 4624     | Logon         |
| Audit Success | 2/23/2021 10:17:31 PM | Microsoft Wind... | 4624     | Logon         |
| Audit Success | 2/23/2021 10:17:20 PM | Microsoft Wind... | 4624     | Logon         |

Event 4624, Microsoft Windows security auditing.

General Details

Subject:

- Security ID: SYSTEM
- Account Name: WIN-CGAK3CTL9KRS
- Account Domain: WORKGROUP
- Logon ID: 0x3e7

Logon Type: 10

New Logon:

- Security ID: WIN-CGAK3CTL9KR\LetsDefendTest
- Account Name: LetsDefendTest
- Account Domain: WIN-CGAK3CTL9KR
- Logon ID: 0x1b3e0ce

Log Name: Security

Source: Microsoft Windows security

Event ID: 4624

Level: Information

User: N/A

OpCode: Info

More Information: [Event Log Online Help](#)

Logged: 2/23/2021 10:17:20 PM

Task Category: Logon

Keywords: Audit Success

Computer: WIN-CGAK3CTL9KR

When we look at the “Logon Type” field, we see the value 10. This indicates that you are logged in with “Remote Desktop Services” or “Remote Desktop Protocol”.

You can find the meaning of the logon type values on Microsoft’s page.

<https://docs.microsoft.com/en-us/windows/security/threat-protection/auditing/event-4624>

In the next section, we will detect the Brute force attack the attacker made before logging in.

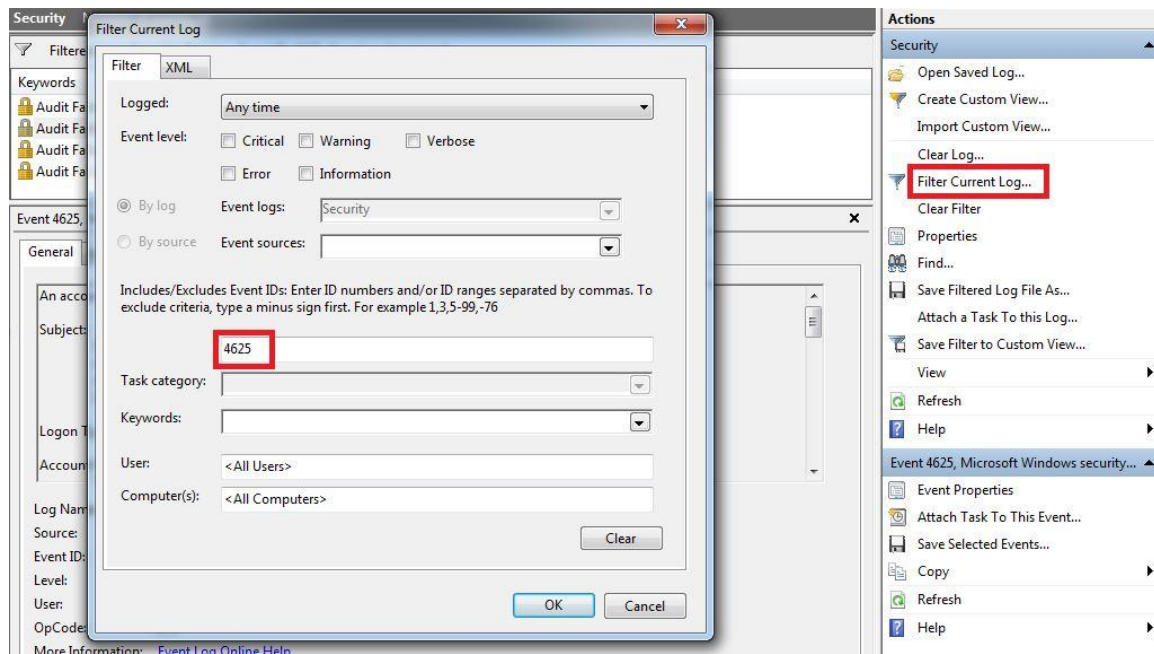
## 7.2 Windows RDP Brute Force Detection

In this section, we will catch an attacker who is in the lateral movement phase. The attacker is trying to jump to the other machine by brute force over RDP.

Download log file: Log\_File.zip Pass=321

[Log\\_File.zip Pass=321](https://files-ld.s3.us-east-2.amazonaws.com/Log_File.zip) ([https://files-ld.s3.us-east-2.amazonaws.com/Log\\_File.zip](https://files-ld.s3.us-east-2.amazonaws.com/Log_File.zip))

When an unsuccessful login operation is made on RDP, the "Event ID 4625 - An account failed to log on" log is generated. If we follow this log, we can track down the attacker.



After filtering, we see 4 logs with 4625 Event IDs.

| Filtered: Log: Security; Source: ; Event ID: 4625. Number of events: 4 |                       |                   |          |               |
|------------------------------------------------------------------------|-----------------------|-------------------|----------|---------------|
| Keywords                                                               | Date and Time         | Source            | Event ID | Task Category |
| Audit Failure                                                          | 2/23/2021 10:17:11 PM | Microsoft Wind... | 4625     | Logon         |
| Audit Failure                                                          | 2/23/2021 10:17:01 PM | Microsoft Wind... | 4625     | Logon         |
| Audit Failure                                                          | 2/23/2021 10:16:58 PM | Microsoft Wind... | 4625     | Logon         |
| Audit Failure                                                          | 2/23/2021 10:16:56 PM | Microsoft Wind... | 4625     | Logon         |

When we look at the dates, we see that the logs are formed one after the other. When we look at the details, it is seen that all logs are created for the "LetsDefendTest" user.

| Keywords      | Date and Time         | Source            | Event ID | Task Category |
|---------------|-----------------------|-------------------|----------|---------------|
| Audit Failure | 2/23/2021 10:17:11 PM | Microsoft Wind... | 4625     | Logon         |
| Audit Failure | 2/23/2021 10:17:01 PM | Microsoft Wind... | 4625     | Logon         |
| Audit Failure | 2/23/2021 10:16:58 PM | Microsoft Wind... | 4625     | Logon         |
| Audit Failure | 2/23/2021 10:16:56 PM | Microsoft Wind... | 4625     | Logon         |

Event 4625, Microsoft Windows security auditing.

|                                                                                                                                                                                                                                                                                                                          |         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| General                                                                                                                                                                                                                                                                                                                  | Details |
| <p>Account For Which Logon Failed:</p> <p>Security ID: NULL SID</p> <p>Account Name: LetsDefendTest</p> <p>Account Domain: WIN-CGAK3CTL9KR</p> <p>Failure Information:</p> <p>Failure Reason: Unknown user name or bad password.</p> <p>Status: 0xc000006d</p> <p>Sub Status: 0xc000006a</p> <p>Process Information:</p> |         |

As a result, we understand that the attacker has unsuccessfully attempted to login 4 times. To understand whether the attack was successful or not, we can search for the 4624 logs we saw in the previous section.

Filter Current Log

Filter XML

Logged: Any time

Event level: ☐ Critical ☐ Warning ☐ Verbose  
☐ Error ☐ Information

☒ By log Event logs: Security

☐ By source Event sources:

Includes/Excludes Event IDs: Enter ID numbers and/or ID ranges separated by commas. To exclude criteria, type a minus sign first. For example 1,3,5-99,-76

4625,4624

Task category:

Keywords:

User: <All Users>

Computer(s): <All Computers>

Clear

OK Cancel

| Filtered: Log: Security; Source: ; Event ID: 4625,4624. Number of events: 14 |                       |                   |          |               |  |
|------------------------------------------------------------------------------|-----------------------|-------------------|----------|---------------|--|
| Keywords                                                                     | Date and Time         | Source            | Event ID | Task Category |  |
| Audit Success                                                                | 2/23/2021 10:17:20 PM | Microsoft Wind... | 4624     | Logon         |  |
| Audit Failure                                                                | 2/23/2021 10:17:11 PM | Microsoft Wind... | 4625     | Logon         |  |
| Audit Failure                                                                | 2/23/2021 10:17:01 PM | Microsoft Wind... | 4625     | Logon         |  |
| Audit Failure                                                                | 2/23/2021 10:16:58 PM | Microsoft Wind... | 4625     | Logon         |  |
| Audit Failure                                                                | 2/23/2021 10:16:56 PM | Microsoft Wind... | 4625     | Logon         |  |

Event 4624, Microsoft Windows security auditing.

|                                                                                                                                                                                                                                                                                                                                                 |         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| General                                                                                                                                                                                                                                                                                                                                         | Details |
| <p>New Logon:</p> <p>Security ID: WIN-CGAK3CTL9KR\LetsDefendTest</p> <p>Account Name: LetsDefendTest</p> <p>Account Domain: WIN-CGAK3CTL9KR</p> <p>Logon ID: 0xd3e0ce</p> <p>Logon GUID: {00000000-0000-0000-0000-000000000000}</p> <p>Process Information:</p> <p>Process ID: 0xd118</p> <p>Process Name: C:\Windows\System32\winlogon.exe</p> |         |

As can be seen from the results, the attacker succeeded in connecting to the system with the 4624 log after the 4625 logs.

Source: Let's Defend.